

Run 2 — Short Analysis

Plasticity → SVAMP-stall probe · Qwen2.5-0.5B-Instruct · 450 GSM8K GRPO updates · constant LR $1e-5$ · $\beta = 0$

Summary: the model collapsed almost immediately, the target regime is a razor-thin slice of hyperparameters, and entropy — not the plasticity metrics — was the earliest warning.

Run 2 did not test our hypothesis, because it never produced the conditions the hypothesis is about. Instead it destroyed the model, and in doing so told us something useful about where the real experiment has to live.

The model failed completely, and fast

Stage A never had a healthy phase. Training reward peaked at the *first* logged step (0.375) and declined from there; by update ~35 the model was producing near-zero reward, roughly 90% of prompt groups had identical rewards (so GRPO gradients were zero), and completions were slamming into the length cap. By update ~235 every output was a 384-token repetition loop at entropy 0.001 — fully deterministic garbage. The final **215 updates (half the run) did nothing**: zero reward, zero gradient, a dead model spinning on the GPU. Greedy GSM8K accuracy confirms it: 0.45 → 0.05 by checkpoint 50 → 0 thereafter. Constant LR $1e-5$ with no KL anchor overshot straight past the useful regime into policy collapse.

The target regime is a razor-thin slice of hyperparameters

Our hypothesis concerns a specific, hard-to-hit state: stage A *succeeding* while capacity quietly erodes underneath. We have now bracketed that state from both sides without landing in it. Run 1 (LR $5e-6$, cosine→0) was too gentle — nothing moved, nothing eroded. Run 2 (LR $1e-5$ constant) was too hot — the model broke before anything could quietly erode. The window where training stays healthy *and* plasticity degrades is narrow, and we have not yet found it. This is worth flagging honestly: if quiet capacity loss during successful training only exists in a thin hyperparameter band, that itself is a modest argument against how common and practically important the failure mode is — a question the proposal should address rather than assume.

The plasticity metrics lagged; entropy led

Where the run does bear on the hypothesis, the news is not good for the headline detector. The hypothesis is that Q (effective rank, dormant fraction) warns *before* the ordinary dashboard signals. It did the opposite. The dashboard signals — collapsing reward, exploding clip ratio, fraction-of-zero-std-groups rising to 0.9 — fired by update ~35. Effective rank was still reading a healthy ~1708 at checkpoint 150, more than 100 updates *after* the model was already at 0% accuracy. Dormant fraction barely moved even after total collapse. In this run Q trailed the failure by 100+ updates; it gave negative lead time.

Entropy is the exception, and the most promising thread across both runs. It **spiked** to ~ 0.59 just before checkpoint 50 — a destabilization signal — right as the damage was happening, then crashed. An entropy *anomaly* (not just its level) preceded the visible collapse. If the plasticity metrics keep trailing while entropy keeps leading, the honest form of this project may be entropy-based stall prediction, with effective rank demoted to a baseline that Q must beat rather than the hero metric. That is exactly the bake-off the proposal already commits to — the detector question doing its job, not a failure of the project.

Where this leaves us

Not a confirmation and not a clean refutation — the decisive test hasn't been run. Status across three data points: too cold, too hot, target regime still unhit. **Run 3** needs to land between: LR $5e-6$ *constant* ($5e-6$ was stable in run 1; constant keeps more pressure than the decayed schedule), $\beta = 0$, plus two safeguards this log makes obvious — a mid-run GSM8K health eval every ~ 50 updates so we can *see* stage A staying alive, and an early-abort guard so a dead run doesn't waste half a session. If $5e-6$ constant is still too hot, the next lever is $1e-5$ with a small KL anchor ($\beta \approx 0.02$). Entropy deviation should be tracked as a first-class predictor from here on, not an afterthought.